Credit, Fundamentals, and the Price–Rent Disconnect in Australian Housing Markets: Evidence from Five Capital Cities, 2006–2025

Abstract

We document a structural change in how dwelling prices and rents are determined across Australia's five largest capital-city markets, and characterise how a common credit impulse reaches each. Using quarterly data for 2006:Q3–2025:Q4, we estimate two reduced-form equations on the within-city time dimension: a rent equation and a price equation. Rents are well explained by market fundamentals throughout the sample, namely a migration-to-completions tightness measure, real wages, and employment, and this fundamentals structure remains the operative one throughout, though the relative weight of its channels shifts over time, with the employment channel strengthening in the second half of the sample. The price equation tells a different story. Investor mortgage demand comoves far more tightly with dwelling prices than with rents, and after 2017 this asymmetry widens. Because investor credit and prices are partly determined within the same quarter, we bound the relationship: the contemporaneous association is large, and it attenuates but does not vanish when credit is entered as predetermined. Panel local projections resolve the timing: a credit innovation is capitalised into prices on impact and reaches rents only with a lag of more than a year, so the two responses converge over a three-year horizon. The decoupling is thus a difference in transmission speed and a medium-run wedge rather than a permanent insulation of rents from financial conditions. The pattern is broad-based across all five capitals rather than concentrated in any single city. We interpret the breakdown of the price–rent relationship as the two markets coming to be governed by different forces: rents by local housing fundamentals, prices increasingly by the cost and availability of investor finance. This has a direct policy implication. Because the two sides of the market respond to different drivers on different timescales, an instrument calibrated to one side, a rental-affordability measure acting on a demand side still anchored to fundamentals, is poorly aligned with a price side that responds to financial conditions; the rent consequences of a credit-driven price boom, in particular, materialise only after a delay. We document this separation rather than estimate the effect of any single policy, and our extended panel does not support a reading in which identical national policy produces a financialization gradient across cities.

Keywords: Regional development, Housing financialization, Rental markets, Price–rent disconnect, Investor credit, Negative gearing, Local projections

1. Introduction

Regional development theory has long emphasized the coordination of economic growth, employment, and social equity within specific territorial contexts (Kuklinski, 1970), assuming that market mechanisms would align investment with regional needs through price signals and competition. The financialization of urban economies has disrupted these assumptions, introducing distortions beyond the reach of traditional policy coordination (Fernandez and Aalbers, 2016; Pike et al., 2019).

Nowhere is this disruption more visible than in rental markets. Across Australia's major cities, between 30% and 51% of renters spend more than 30% of their income on rent, with

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the most financialized markets recording the highest stress rates. This reflects a global pattern in which market deregulation has produced rental housing that is costly, insecure, and often substandard (Bricocoli and Peverini, 2024; McKee et al., 2017; Waldron, 2023), ultimately undermining regional economic competitiveness by pricing out essential workers.

The underlying driver is the transformation of housing from shelter into financial asset (Stephens, 2024), fuelled by mortgage expansion and tax incentives such as negative gearing (Vidal et al., 2024). When investors prioritise capital gains over rental income, the traditional supply–price relationship weakens and policy effectiveness erodes (Hulse et al., 2020). Housing supply may increase yet rental security deteriorate, as short-term leases, rent hikes, and asset speculation displace the provision function (Aalbers et al., 2021; Fields, 2017).

Yet a critical gap remains: the extent to which these dynamics vary *across* metropolitan markets within a single national system. Existing evidence draws almost entirely on single-city case studies, leaving unanswered whether financialization produces a uniform national distortion or a gradient of outcomes that depends on local investor concentration, institutional structures, and geographic constraints. Answering this question requires comparative metropolitan analysis within a shared policy environment.

Australia's five largest state capital cities provide an ideal setting for this investigation. They share a common national policy framework, including monetary policy, negative gearing, and migration settings, but differ substantially in investor concentration, geographic constraints, and migration exposure. Sydney and Melbourne, the most heavily invested markets, face severe geographic limitations and high investor concentration; Brisbane and Perth represent rapid-growth markets with distinct housing cycles; Adelaide offers a less constrained comparator with lower investor penetration. These shared national settings, with differing local conditions, make the five capitals a natural panel in which to ask how rent and price determination evolved over time within each market.

Three features make this setting particularly relevant. First, Australia's negative gearing system fosters a distinctive investor landscape, which Hulse et al. (2020) characterizes as "mum and dad" financialization, setting it apart from institutional models in other countries. Second, the federal structure generates multi-scalar policy interactions, with national monetary and tax settings intersecting state-level housing reforms. Third, the 2017 NSW Housing Affordability Package provides a concrete, dateable institutional event: announced in December 2016, the package expanded community housing by 40.7% while reducing public housing by 7.0%, restructuring the institutional delivery of social housing in Australia's most financialized market. It gives a specific break date around which to ask whether the determinants of prices and rents changed; we treat it as one marker within a longer shift rather than as a treatment whose causal effect we estimate (Section 3.3).

Research Questions and Approach. We ask a structural question rather than a policy-evaluation one. What drives rents, and what drives prices, in these five markets; and did those drivers change over our sample? Specifically: (1) Are rents and dwelling prices governed by the same forces, or by different ones? (2) Did the determinants of either shift around the mid-2010s, when the 2017 NSW package was introduced and investor activity in housing intensified nationally? (3) If prices and rents decoupled, is that because one of them migrated onto a distinct set of drivers?

Our empirical strategy places inferential weight where a five-city quarterly panel can bear it: on the within-city time dimension, which contains roughly seventy quarters of variation per market, rather than on the thin five-unit cross-section. We estimate two reduced-form equations. A rent equation relates rent growth to a market-tightness fundamental (net migration relative to dwelling completions), real wages, and employment. A price equation relates price growth to the cost of capital (the cash rate) and investor mortgage demand, alongside the same fundamentals. We then test whether the financial driver in each equation changed after 2017 by interacting investor credit with a post-2017 indicator, and we trace the timing with a rolling-window decomposition of the share of variation each equation attributes to investor demand. The design is descriptive: each equation is a standard reduced form, and we make no claim to identify the causal effect of any single policy. This is deliberate. It answers the natural "correlation, not causation" objection by not overclaiming causation, and it identifies the result off the dimension of the data that is actually informative.

Contributions. This paper makes three contributions. First, we show that rents and dwelling prices in these markets are governed by different forces: rents track local fundamentals (market tightness, real wages, and employment) throughout the sample, holding across cities, while prices respond to the cost of capital and to investor credit. The breakdown of the historical price–rent relationship follows directly from this, rather than being an unexplained correlation.

Second, we characterise the differing role of investor credit in the two markets, and we are careful about what the comparison can bear. Investor credit comoves far more tightly with prices than with rents, and this asymmetry widens after 2017. Because credit and prices are partly co-determined within the quarter, we bound the relationship rather than report a single elasticity: the contemporaneous association is large, and it attenuates but does not vanish when credit is entered as predetermined. The rent equation serves as a placebo; if the price–credit association were a pure artifact of joint determination, the same force would not so largely spare rents.

Third, we characterise the timing of the contrast using panel local projections. A credit innovation is capitalised into prices on impact, consistent with an asset-pricing channel, and reaches rents only with a lag of more than a year, at the slower speed of occupancy and supply; over a three-year horizon the two responses converge. The disconnect is therefore a difference in transmission speed and a medium-run wedge rather than a permanent insulation of rents from financial conditions. We further show the pattern is present across the five capitals and not concentrated in Sydney, the city targeted by the 2017 package, and that its sharpest movements are centred on the pandemic period, so we treat the 2017 reform as one marker within a longer structural shift rather than as its cause. A genuinely powered cross-sectional test of a financialization gradient would require sub-metropolitan data, which we flag as a natural extension.

The remainder of the paper proceeds as follows. Section 2 reviews the literature; Section 3 describes data and methods; Section 4 presents results; Section 5 discusses implications; and Section 6 concludes.

2. Literature and Theoretical Background

The paper sits within housing economics and draws on three economic literatures, the user-cost theory of house prices, the supply-constraint and monetary-transmission literatures, and the migration–housing nexus, before turning to the financialization literature that describes the phenomenon we explain. We position the contribution against the economics of housing first, and treat the broader financialization account as the description of the decoupling whose mechanism we identify.

The organising idea is the standard one in housing economics: a dwelling is at once a consumption good, whose rent is the price of housing services, and an asset, whose price is the present value of those services and of expected capital gains. In the user-cost framework the two are linked through the cost of capital, so that prices and rents move together when both respond to broad housing demand and can diverge when the asset side loads on financial conditions that the service side does not. This is the link our estimation tests directly, by asking whether prices and rents have come to load on different drivers. The cointegration literature on housing provides the time-series counterpart, treating rents and prices as potentially I(1) series whose long-run relationship can weaken (Kenny, 1999); we take the question of whether that relationship has broken down to the within-city panel.

On the supply side, Glaeser and Gyourko (2003) established that regulatory and geographic constraints shape regional housing supply, but more recent work finds that financial factors increasingly dominate the supply response. Greenaway-McGrevy and Phillips (2023) show that

Auckland's land-use deregulation yielded minimal affordability gains because investor demand, driven by capital-gains expectations rather than rental yields, absorbed the new supply, and Egan and McQuinn (2023) identify regime shifts in which the traditional supply–demand relationship breaks down, so that interventions premised on classical market assumptions can fail precisely when they are most needed. These findings motivate a design in which the price response to investor finance is allowed to differ from the rent response to fundamentals rather than being constrained to move with it.

The assumption of uniform national policy effects is itself challenged by evidence of regionally differentiated monetary transmission. Koeniger et al. (2022) document how institutional differences across European regions cause identical central-bank actions to produce divergent housing outcomes, and Calza et al. (2013) show that monetary policy disproportionately affects regions with more volatile employment. In Australia, Graham and Read (2023) find that interest-rate changes affect metropolitan and regional markets very differently. A recognised gap, noted by Stephens (2024), is that this literature has concentrated on homeownership and the mortgage channel and has relatively little to say about rental-market dynamics specifically (Kuang et al., 2026), even as rental housing accommodates a growing share of urban populations. Separating the rent and price equations and tracing how a common credit impulse reaches each is one way to address that gap.

Migration enters as a demand fundamental on the rent side, and the economics literature here is uneven. Work on immigration and house prices is well developed (Saiz, 2010; Sanchis-Guarner, 2023), but the effect on rents is less studied despite renters' over-representation among recent migrants (Nygaard, 2011). Sharpe (2019) attribute a substantial share of US rental price variation to immigration, and Moallemi and Melser (2020) find that immigration effects on Australian rents vary across cities, with Sydney most exposed. Li et al. (2024) add that housing costs themselves drive internal migration in Australia, so that constrained markets export population to less constrained ones, a sorting channel through which rents feed back into the regional distribution of labour (Haas and Osland, 2014). We capture the demand side of this nexus through a market-tightness measure, net migration relative to dwelling completions, rather than through migration alone.

Against this economic background, a large literature documents the transformation of housing from shelter into financial asset (Aalbers et al., 2021; Fernandez and Aalbers, 2016) and the resulting decoupling of prices from rents. The form of this process is known to vary with local institutions (Christophers, 2015); in Australia Hulse et al. (2020) describe a distinctive small-investor financialization driven by negative gearing. This literature establishes the phenomenon, prices pulling away from rents, but typically leaves the decoupling as a documented correlation rather than a mechanism. We take it in a different direction. Rather than asking whether the degree of financialization varies across markets, we ask whether it drives a wedge *within* a market, between the determinants of prices and the determinants of rents, and we trace how a common financial impulse reaches each outcome. This reframes the decoupling as the mechanical consequence of the two outcomes loading on different drivers, which the within-city panel can test.

Two features distinguish our design from this prior work. First, the within-market question, whether the determinants of rents and prices diverge over time, has not been posed: Moallemi and Melser (2020) compare migration–rent dynamics across the Australian capitals but do not examine the broader transmission channels or the divergence of price and rent determination. Second, identifying that divergence requires a common policy environment with within-market time variation rather than a single-city study, which Koeniger et al. (2022) show is the informative dimension for separating systematic transmission from city-specific idiosyncrasy. The five Australian capitals provide exactly this setting.

Our contribution synthesises these literatures. Estimating separate rent and price equations on the within-city time dimension across five Australian metropolitan markets, we show that the two outcomes come to be governed by different forces, fundamentals for rents and investor finance for prices, and we trace the dynamic mechanism, fast capitalisation into prices and slow transmission to rents, that produces the breakdown of the price–rent relationship.

3. Methods

3.1. Conceptual Framework

Our starting point is that a dwelling is two things at once: a place to live, whose rent is the price of a housing service, and an asset, whose price is a claim on future housing services and capital gains. When both are governed by the same broad housing demand, rents and prices move together, which is the historical norm in these markets. They can come apart if the two outcomes begin to load on different drivers, with rents tracking the fundamentals of the service market while prices track the conditions of asset finance. Documenting whether and when that happens, and characterising how a common financial impulse reaches each outcome, is the object of the paper.

This framing yields a small set of expectations that map directly onto the estimation, without requiring us to take a full structural model to the data. We set the underlying supply-and-demand system, including the crowding-out term, the rental-demand split between traditional and finance-driven components, and a policy-misalignment condition, in Appendix A; here we carry only what motivates the two estimating equations and the reduced form they specialise.

The reduced-form rental price equation for city j is

$$R_{jt} = F(C_{j,t-1},\ r_{t-1},\ CO_{j,t-1},\ P_{j,t-1},\ M_{j,t-1},\ w_{j,t-1},\ N_{j,t-1},\ S^G_{jt},\ NH_{jt}), \tag{1}$$

a function of construction costs, the national interest rate, a crowding-out term capturing competition for construction capacity, lagged property prices, migration, wages and employment, public housing supply, and owner-occupier demand. The price equation is the asset-side counterpart, in which the cost of capital and investor mortgage demand enter directly. The estimation retains the parsimonious subset of each that a within-city first-difference design can identify, which we set out in Section 3.4.

Four expectations follow. First, rents are governed by service-market fundamentals, the balance of migration against new dwelling supply (market tightness), wages, and employment, and this set should remain the governing one throughout the sample and hold city by city, even if the relative weight on each channel shifts over time. Second, dwelling prices are governed by the cost of capital and by investor mortgage demand, with at most a secondary role for the rental fundamentals. Third, the loadings of the two outcomes on investor finance should move apart over the sample: after the mid-2010s prices should load more heavily on investor credit while rents should not, which is what produces the observed disconnect. Fourth, because prices capitalise financial conditions quickly while rents adjust through slower occupancy and supply channels, the two should differ in the speed with which an investor-credit innovation reaches them, converging only over the medium run. We do not advance a cross-city financialization gradient as an expectation: with five units it cannot be identified, and we treat any cross-city differences descriptively. We use the pre-2017 property–rental correlations, near-uniform across the capitals (range 0.941–0.979), only to note that the two markets began the sample closely integrated everywhere, not as a cross-sectional moderator.

Figure 1 organises these channels into a rent side tied to fundamentals and a price side tied to investor finance, the separation of which the estimation then tests in the time dimension. Federal settings (monetary policy, negative gearing, migration) shape structural conditions and interact with state-level reforms through market dynamics rather than explicit coordination; the 2017 NSW package enters as one dateable marker within that environment rather than as a treatment whose effect we estimate.

To operationalise these expectations, we specify two reduced-form equations, one for rents and one for prices, and a test that interacts investor demand with the post-2017 period, allowing

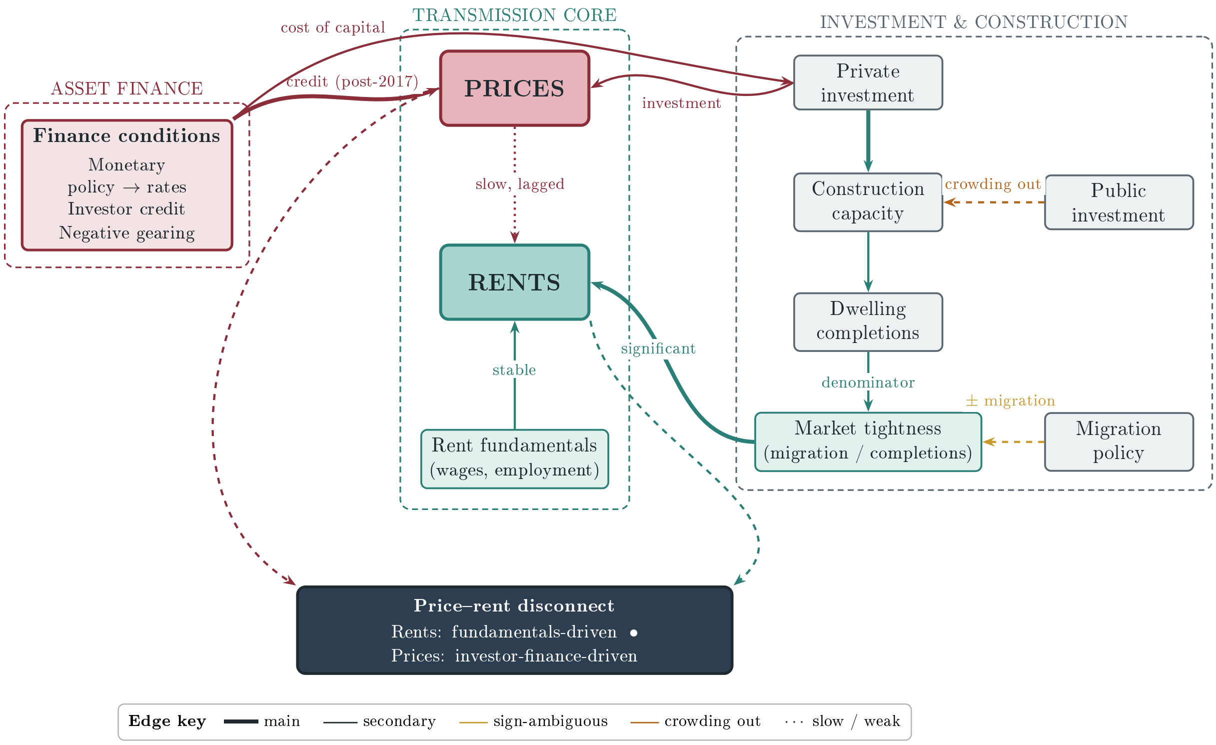

Figure 1: Transmission channels into prices and rents. The asset-finance channels enter on the left: investor credit, the main channel and the one that strengthens after 2017, alongside negative gearing and the cost of capital. The cost-of-capital channel operates through investment, since monetary policy sets interest rates, which move private investment in the investment-and-construction band on the right. Private investment is the hinge between the two sides; it raises prices directly, the fast asset channel, and it builds construction capacity and hence dwelling completions, the slow supply channel, with public investment competing for the same capacity through crowding out. Completions, the denominator, and migration, the numerator, form market tightness, which together with wages and employment drives rents through the transmission core in the middle. A credit impulse thus reaches prices on impact and rents only slowly, through building and through the dotted price-to-rent link, and both outcomes feed the price–rent disconnect. Edge weight and style indicate each channel's role; see the key. The structural model is given in Appendix A.

the data to reveal which channels are active and whether the price–credit relationship strengthens after the break.

3.2. Empirical Specification

Building on the theoretical framework, we estimate a reduced-form rental price equation (Equation 1) as an autoregressive distributed lag (ARDL) model for city j:

$$R_{jt} = \beta_{0j} + \alpha_{1j} R_{j,t-1} + \alpha_{2j} \text{POST}_t + \mathbf{X}'_{jt} \boldsymbol{\beta}_j + \varepsilon_{jt} \tag{2}$$

Here, R_{jt} denotes the rental price index in city j; $\text{POST}_t = \mathbb{1}[t \geq 2017{:}Q1]$ for all cities. This is a policy intervention dummy capturing city-specific housing reforms, which, for non-Sydney cities, captures the post-2017 institutional environment broadly, while in Sydney it is related to the public housing reform. We denote a Sydney-only NSW-reform indicator as $D^{NSW}_{jt} = \mathbb{1}[j = \text{Sydney}] \cdot \text{POST}_t$ and $\mathbf{X}_{jt}$ is a vector of explanatory variables derived from the theoretical model. Specifically, $\mathbf{X}_{jt} = [C_{j,t-1},\ r_{t-1},\ GFCF^P_{j,t-1},\ GFCF^G_{j,t-1},\ P_{j,t-1},\ M_{j,t-1},\ w_{j,t-1},\ N_{j,t-1},\ HHS_{jt}]$. In this specification, $C_{j,t-1}$ refers to construction costs; r_{t-1} to the lagged cash rate (national); $P_{j,t-1}$ is the property price index; $M_{j,t-1}$ captures net migration flows; $w_{j,t-1}$ and $N_{j,t-1}$ denote average wages and employment; and we proxy owner-occupier demand, NH_{jt}, using household housing services consumption (HHS_{jt}). To model the crowding-out effect, the concurrent competition between public and private investment for construction capacity, the theoretical specification includes public and private Gross Fixed Capital Formation ($GFCF^G_{j,t-1}$ and $GFCF^P_{j,t-1}$) separately rather than using the crowding-out ratio $CO_{j,t-1}$ directly, which would in principle allow the data to determine whether the crowding-out mechanism operates differently across cities with varying geographic constraints. In estimation we retain the parsimonious specifications below, which drop the GFCF terms; we return to this reduction where the rent equation is introduced (Section 3.4).

We estimate the two reduced-form equations as within-city first-difference models over 2006:Q3–2025:Q4, with the stationary tightness measure in levels (lagged) and the $I(1)$ series in differences. A mean-group estimator allows fully heterogeneous city-level coefficients in the rent equation, confirming that the fundamentals relationship holds market by market. Identification rests on the within-city time dimension rather than the five-unit cross-section.

3.3. Data Sources and Description

Our empirical design uses quarterly data from five Australian capital cities (Sydney, Melbourne, Brisbane, Adelaide, and Perth) covering 2006:Q3 to 2025:Q4, with most series extending to 2025:Q4 and the binding endpoint set by the publication lag of net migration. The within-city time dimension carries the identification: about seventy-eight quarters of data per market, roughly seventy after differencing and lagged terms enter the estimation, estimated jointly across the five capitals. We document below three corrections to the price and flow series relative to earlier versions of this panel, together with the construction of a continuous price index after the discontinuation of the ABS capital-city series in 2021.

The panel window is determined by binding data constraints: capital city property price indices from the Australian Bureau of Statistics (ABS) Catalogue 6416.0 begin in 2003:Q3, while quarterly state-level overseas migration data begin in 2006:Q3. The five cities were selected based on three criteria: population size (each exceeds one million residents), data completeness across all variables, and variation in housing market structure, spanning highly financialized markets (Sydney, Melbourne), rapid-growth markets (Brisbane, Perth), and a less constrained benchmark (Adelaide).

3.3.1. Variable Construction and Sources

Table 5 (Appendix) provides full variable definitions, sources, and geographic coverage. Briefly, outcome variables are capital-city rental price indices (ABS 6401.0) and residential property price indices (ABS 6416.0); investment is measured by public and private Gross Fixed Capital Formation (GFCF; ABS 5220.0 State Accounts); labour market conditions by employment (ABS 6202.0), wages measured by the Wage Price Index (WPI; ABS 6345.0), and labour income (ABS 5220.0); investor mortgage demand by the value of new investor housing loan commitments by state (ABS Lending Indicators, the dataflow successor to Catalogue 5601.0 Table 14), with national investor and owner-occupier housing credit and the net loan-purpose switching flow from the Reserve Bank's D2 series used in the appendix robustness checks; migration by net overseas and interstate flows (ABS 3101.0), with net overseas migration disaggregated by visa-and-citizenship group (ABS Overseas Migration) employed in the appendix mechanism check; and monetary policy by the cash rate (RBA). State-level variables serve as city proxies, appropriate given that each capital city accounts for 65–79% of its state's economic activity.

Two features of the outcome series warrant explicit statement. The rent measure is the housing (rent) component of the ABS Consumer Price Index (Catalogue 6401.0), an index of rents actually paid rather than a hedonic or repeat-lease rental price index; it is the longest consistent capital-city rent series available and is standard in Australian housing work, but it tracks average rents paid across the stock, not new-lease asking rents, and we interpret it accordingly. For dwelling prices, the ABS discontinued its established residential property price index, so we construct a single continuous capital-city series by splicing: for Sydney we use the Bank for International Settlements all-dwellings series, and for the other four capitals we extend the ABS 6416.0 index with the ABS Total Value of Dwellings mean-price series, chain-linking at the splice point so that growth rates rather than levels are joined. The splice affects the level of the index, which city fixed effects and first-differencing absorb, not the quarter-to-quarter growth that the estimation uses. Table 5 lists each underlying series and its coverage.

3.3.2. Cross-City Summary Statistics

Table 1 presents summary statistics across the five cities for the panel period (2006:Q3–2025:Q4), describing the range of housing-market conditions in the panel.

Table 1: Cross-City Summary Statistics, 2006:Q3–2025:Q4

	Sydney	**Melbourne**	**Brisbane**	**Adelaide**	**Perth**
Price Dynamics					
Rental Index, mean	108.6	106.8	106.0	107.5	98.9
Property Price Index, mean	148.3	131.0	128.5	125.2	113.7
Prop.–Rent Corr. (pre-2017)	0.98	0.97	0.97	0.97	0.95
Prop.–Rent Corr. (post-2017)	0.77	0.26	0.96	0.98	0.97
Correlation change	−0.21	−0.71	−0.01	+0.02	+0.02
Migration (thousands, quarterly mean)					
Net Overseas	19.9	17.7	9.9	3.6	7.5
Net Interstate	−5.0	0.0	4.8	−0.7	0.9
Investment (AUD bn, quarterly mean)					
Public GFCF	7.25	5.31	5.25	1.41	2.44
Private GFCF	25.60	21.35	18.24	5.04	15.93
Public/Private ratio	0.28	0.25	0.29	0.28	0.15
Labour Market					
Employment (millions)	3.81	3.10	2.45	0.84	1.34
WPI Index, mean	123.8	124.6	124.3	124.2	124.3
Market Structure					
Post-2017 corr. decline[b]	Small	Large	None	None	None
Geographic constraint[c]	Severe	Moderate	Low	Low	Moderate

Notes: Up to $N = 78$ quarters per city (binding endpoint 2025:Q4). Indices use 2011–12 base. Rolling correlations use a 20-quarter (5-year) trailing window; pre-2017 and post-2017 values are point-in-time snapshots from the rolling series (not period averages): specifically, the rolling correlation at the last quarter before 2017:Q1 and at 2025:Q4 respectively. [b]Large = post-2017 rolling correlation decline > 0.30; Small = between 0 and 0.30; None = no decline. Reported descriptively, not as a financialization-intensity index. [c]Following Saiz (2010): Severe = multiple natural barriers; Moderate = partial; Low = unconstrained.

Three patterns in Table 1 describe the variation in the panel. First, the post-2017 decline in the property–rental correlation is largest in Sydney and Melbourne and small or absent in the other capitals; we report this as a descriptive contrast, not as an identified gradient (Figure 2). Second, migration pressures are unevenly distributed: Sydney and Melbourne receive large overseas inflows but experience net interstate outflows, consistent with affordability-driven displacement (Li et al., 2024), whereas Brisbane receives both overseas and interstate migrants. Third, the public-to-private GFCF ratio and absolute investment magnitudes differ substantially, which is relevant to whether the crowding-out mechanism operates differently across capacity-constrained and unconstrained cities.

Table 2 reports summary statistics for the variables as they enter the estimation: the $I(1)$ series in quarter-on-quarter log-differences, market tightness in levels, and the price–rent gap in log levels. The differenced series are small and roughly symmetric, which is the behaviour the log-difference transformation is designed to produce; tightness, a ratio of net migration to completions, is the one variable with a wider spread, and it takes occasional negative values in the 2020–21 border-closure quarters when net migration itself turned negative.

Index variables (rents, the wage price index) are expressed on a common base and are comparable in level across cities, while the level variables differ substantially with city size. These level differences are absorbed by city fixed effects and first-differencing, so the estimation works off within-city changes rather than cross-city levels. The large cross-city dispersion in property prices, with Sydney highest, reflects city size and accumulated price growth rather than anything the design needs to balance.

Table 2: Summary Statistics for Estimation Variables, 2006:Q3–2025:Q4

Variable	N	Mean	SD	Min	Max
Δ ln Rent	385	0.008	0.009	−0.025	0.033
Δ ln Price	385	0.013	0.024	−0.057	0.092
Δ ln Investor credit	385	0.014	0.148	−0.326	0.394
Δ ln WPI (wages)	385	0.007	0.004	−0.001	0.023
Δ ln Employment	385	0.005	0.010	−0.066	0.043
Δ Cash rate	385	−0.030	0.409	−1.887	1.393
Tightness (level)	370	1.430	0.931	−1.354	5.497
Price–rent gap (log level)	390	0.172	0.169	−0.132	0.669

Notes: Pooled across the five capitals. Δ ln denotes the quarter-on-quarter log-difference (a value of 0.010 is approximately one percent growth); Δ Cash rate is the quarter-on-quarter change in percentage points. Tightness is net migration over the trailing four quarters relative to dwelling completions over the same window, entered in levels and lagged one quarter in estimation; its negative minimum reflects the 2020–21 border-closure quarters in which net migration was negative. The price–rent gap is the log of the dwelling-price index over the rent index, entered as a lagged level control. N is the number of city-quarters available for each variable after differencing and lagging; tightness loses further observations to its trailing four-quarter construction and one-quarter lag.

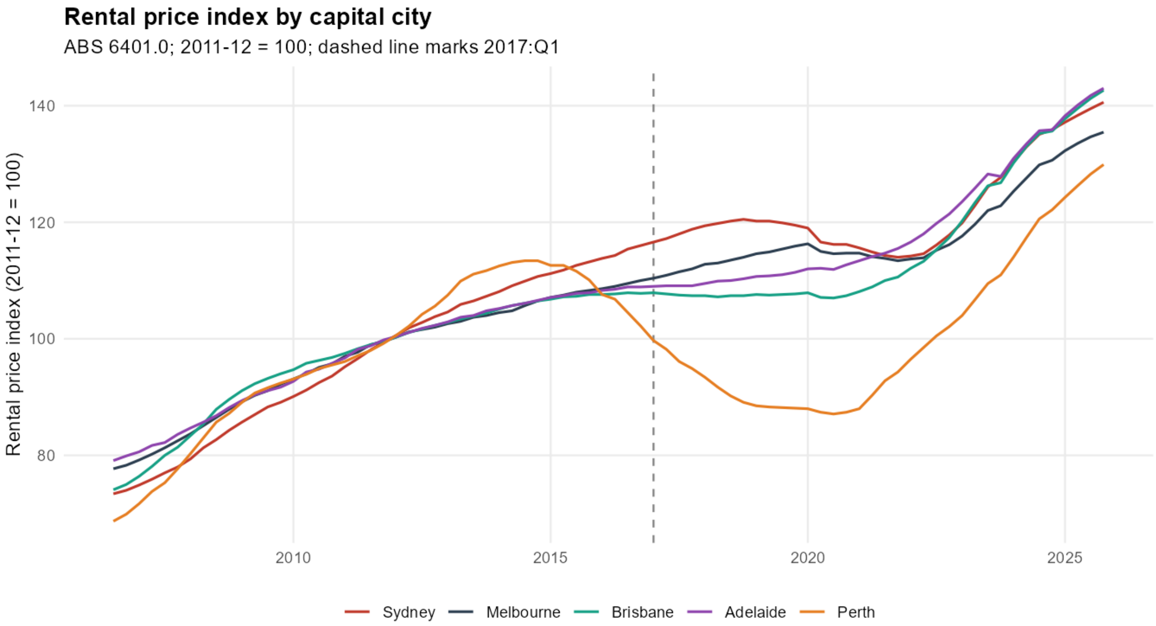

Figure 2: Rental price index trajectories, five Australian capital cities, 2006:Q3–2025:Q4 (ABS 6401.0; base: 2011–12 = 100). The dashed line marks 2017:Q1. Perth diverges from ∼2014 with the mining investment cycle. The series move broadly together, consistent with rents across the capitals being driven by common housing-demand fundamentals; the figure is descriptive and does not underpin a treatment-control comparison.

3.3.3. The 2017 Break Date

We set the post-2017 indicator to begin in 2017:Q1, the quarter after the December 2016 announcement of the NSW package (Section 1), fixing the break date from the institutional timeline. The indicator marks a candidate break in the sample, not a treatment whose effect we estimate; the analysis tests whether the determinants of prices and rents change around it.

The rolling price–rent correlation declines after the mid-2010s across the panel, with the largest declines recorded in Sydney and Melbourne (Figure 3, Section 4.6). The decline is broad-based rather than specific to Sydney, which is consistent with common macroeconomic and institutional forces operating across the metropolitan system rather than with a change confined to one city. We report these correlation patterns as description only. Our estimation (Section 4) finds the post-2017 strengthening of the price–investor-credit channel present across the capitals rather than confined to the most expensive markets; if anything it is weaker in Sydney. The cross-city differences in the measured magnitude of the decoupling are therefore not a basis for categorical claims about differential financialization.

3.3.4. Data Limitations

Three limitations warrant acknowledgement. First, state-level variables serve as proxies for city-level measures for GFCF, labour income, and housing services consumption. While capital cities dominate their state economies, this introduces measurement error that may attenuate estimated coefficients. Second, the panel begins in 2006:Q3, precluding analysis of pre-GFC financialization dynamics; identification rests on the within-city time variation over 2006–2025 rather than on the pre-GFC period. Third, construction cost data at the city level are proxied using state-level implicit price deflators from the national accounts, which may not fully capture city-specific cost pressures.

3.4. Econometric Methods

We estimate two reduced-form equations on the five-city quarterly panel, placing identification on the within-city time dimension. Most series are $I(1)$ index levels, so we work in first differences; the market-tightness measure is a ratio of flows and is stationary, so it enters in levels lagged one quarter. All models include city fixed effects and use heteroskedasticity- and autocorrelation-robust (Driscoll–Kraay) standard errors.

Each method below has a specific and limited role, and the set is small by design. The two reduced-form equations, estimated by within-city fixed effects, establish what governs rents and what governs prices. The mean-group estimator checks that the rent relationship holds city by city rather than only on average. The divergence test, a single interaction of investor credit with a post-2017 indicator in each equation, asks whether the price–credit relationship strengthens after the break while the rent–credit relationship does not. The local projections recover the timing that the static interaction compresses into one coefficient, showing how fast a credit innovation reaches prices versus rents. The shift-share instrument in the appendix is a supplementary check on the price–credit association, reported as indicative given the small cross-section, not as the headline. Everything else is robustness. We do not use the methods to layer additional claims; each addresses one link in a single argument, from what drives each outcome, to whether the credit link changes after 2017, to how quickly it transmits.

Stationarity and functional form. Panel and city-level unit-root tests indicate that log rents, log prices, log wages, log employment, and log investor commitments are $I(1)$, consistent with the cointegration literature on housing (Kenny, 1999). We therefore difference these series. Market tightness, net migration over the trailing four quarters relative to dwelling completions over the same window, is stationary (its autoregressive root is well below unity), and entering it in levels rather than differences is both the correct treatment of an $I(0)$ regressor and materially better-fitting: differencing an already stationary variable over-differences it. The price–rent gap enters the rent equation in lagged log levels as a level control rather than as an error-correction term: a residual-based panel test does not find robust cointegration between log rents and log prices (group-mean ADF ≈ -1.86, around the Pedroni five percent critical value), and the gap carries a coefficient close to zero and statistically insignificant in every rent specification we estimate, so we do not impose an error-correction interpretation on it.

The estimated rent equation is a parsimonious reduction of the theoretical reduced form in Equation 1. We retain the terms the within-city first-difference design can identify and the data support, the tightness fundamental, wages, employment, and the lagged price–rent gap as a level control, and drop those that enter weakly or are absorbed by the transformation: construction costs and the cash rate carry no direct role on the rent side once prices are differenced, the public and private GFCF terms that proxy crowding-out are a supply-side channel that the parsimonious rent and price specifications do not retain, and owner-occupier demand is collinear with the tightness and price-gap terms. The structural model in Appendix A sets out the fuller channel set, including crowding-out as competition for construction capacity; Equations 3 and 4 estimate the subset that survives, and Figure 1 shows the thesis those two equations test.

Rent equation.

$$\Delta r_{it} = \alpha_i + a\,(p-r)_{i,t-1} + b_1\,\text{TIGHT}_{i,t-1} + b_2\,\Delta w_{it} + b_3\,\Delta n_{it} + \varepsilon_{it}, \tag{3}$$

where r is log rent, $(p-r)$ the log price–rent gap, TIGHT market tightness, w log wages, and n log employment. We estimate Equation 3 both as a pooled within (fixed-effects) estimator and as a mean-group estimator (Pesaran and Smith, 1995) that averages city-specific regressions, the latter confirming that the fundamentals relationship holds market by market rather than only on average.

Price equation.

$$\Delta p_{it} = \gamma_i + c_1\,\Delta\text{CASH}_t + c_2\,\Delta\iota_{it} + c_3\,\text{TIGHT}_{i,t-1} + c_4\,\Delta w_{it} + \eta_{it}, \tag{4}$$

where p is log price, CASH the cash rate, and ι log investor mortgage commitments.

Divergence test. The central test interacts investor demand with a post-2017 indicator POST_t in both equations:

$$\Delta y_{it} = \delta_i + \pi_1\,\Delta\iota_{it} + \pi_2\,(\Delta\iota_{it}\times\text{POST}_t) + \mathbf{x}'_{it}\boldsymbol{\psi} + \xi_{it}, \tag{5}$$

estimated separately with $y = p$ and $y = r$ and the appropriate controls $\mathbf{x}_{it}$ from Equations 4 and 3. The cash rate enters in differences but is not interacted: across the post-2017 window it is dominated by the COVID near-zero-rate period and the 2022–23 hiking cycle, making its interaction a regime artifact rather than a financialization signal; we report the cash-rate-interacted version as a robustness check. The parameter of interest is π_2. A large π_2 in the price equation with at most a small π_2 in the rent equation indicates that investor credit comoves more tightly with prices than with rents, and increasingly so after 2017. Because $\Delta\iota_{it}$ and Δp_{it} are partly determined within the quarter, we estimate Equation 5 both contemporaneously and with investor credit entered as predetermined (lagged one quarter); the two bracket the relationship, the contemporaneous estimate as an upper bound that absorbs simultaneity and the predetermined estimate as a lower bound that displaces a within-quarter relationship. The rent equation serves as a placebo for reverse causality. We test $\pi_2 = 0$ in each equation with a one-degree-of-freedom Wald test, and as a further check instrument $\Delta\iota_{it}$ with a shift-share instrument (Appendix C.1), reading the first-stage strength as the binding diagnostic given the small cross-section.

Local projections. To recover the dynamics that the static interaction compresses into a single coefficient, we estimate Jordà panel local projections of the price and rent levels on an investor-credit innovation over horizons $h = 0, \ldots, 12$, with city fixed effects and lagged controls that absorb the predictable component of credit. The impulse is scaled to one standard deviation of credit growth. Horizon zero retains the within-quarter simultaneity caveat; horizons from one quarter out are predetermined and clean of reverse causality. The object of interest is the difference in the shape and timing of the two response paths.

4. Results

We organise the results around the two equations and their divergence. Section 4.1 states the identification strategy and its scope. Section 4.2 establishes that rents are governed by fundamentals, that this holds across cities, and that the fundamentals remain the operative drivers throughout even as their relative weights shift over time. Section 4.3 shows that prices respond to the cost of capital and to investor credit. Section 4.4 presents the central contrast and is careful about what it is: a differential co-movement of investor credit with prices and rents, large contemporaneously and attenuating but not vanishing once credit is predetermined. Section 4.5 recovers the dynamic mechanism behind that contrast through local projections, showing that credit reaches prices fast and rents slowly. Section 4.6 reports the cross-city decoupling as description.

4.1. Identification and Scope

Our design is descriptive reduced-form, and we are explicit about what it can and cannot deliver. The two equations are standard reduced forms estimated off roughly seventy quarters of within-city variation, which is where the statistical power in a five-city panel resides. We do not claim to identify the causal effect of any single policy on rents; the 2017 NSW package enters as a structural-break date, one event within a broader shift, not as a treatment whose effect we estimate. Correspondingly, the cross-city dimension is too thin (five units) to identify a financialization gradient, and we use it descriptively rather than as the basis for a typology. This is a deliberate reallocation of inferential weight onto the part of the data that can bear it: the changing time-series structure of rent and price determination within cities, and away from the single-unit and cross-sectional comparisons that cannot. A genuinely powered cross-sectional test of the financialization gradient would require sub-metropolitan (SA4/SA3) data; we flag this as a natural extension rather than attempt it here.

We estimate both equations as within-city (fixed-effects) first-difference models. The $I(1)$ index series (log rents, log prices, log wages, log employment, log investor commitments) enter in differences; the market-tightness measure, a ratio of flows that is stationary, enters in levels lagged one quarter. Standard errors are heteroskedasticity- and autocorrelation-robust (Driscoll–Kraay). We report a mean-group estimator alongside the pooled within estimator for the rent equation to confirm that the relationship holds city by city.

4.2. Rents are governed by fundamentals

Table 3 reports the rent equation. Rent growth responds strongly and significantly to the lagged market-tightness ratio, to wage growth, and to employment growth, and these relationships are stable across cities: the mean-group estimates, which average city-by-city regressions, recover the same pattern as the pooled within estimates. The lagged price–rent gap does not enter rent growth in either specification, with a point estimate close to zero and a p-value of 0.67 in the pooled model and 0.18 in the mean-group model. Rents track the fundamentals that a standard housing-demand account would predict, and they do not chase the widening gap between prices and rents. This result is robust across every specification we report below and is the empirical anchor of the paper.

The fundamentals structure is stable across cities, but its internal weights are not constant over time, and we are explicit about this. A Chow-style test that interacts all four rent regressors with the post-2017 indicator rejects coefficient equality ($\chi^2_4 = 18.7$, $p < 0.001$), so the rent equation is not a fixed-coefficient relationship across the sample. The pattern behind the rejection is interpretable rather than destabilising: interacting the fundamentals with a linear time trend, the employment channel strengthens over the sample (trend interaction $+0.013$, $p = 0.06$, with the rolling-window employment coefficient rising from near zero in the early 2010s to about 0.15 by the 2020s), while the wage and tightness coefficients remain positive and significant throughout but with a noisier wage pass-through. The substantive claim is therefore that rents remain governed by the same service-market fundamentals across the whole sample, with the relative weight on employment growing as the labour market tightened; it is not that the coefficients are literally constant. This evolution does not affect the divergence result, which concerns the investor-credit channel, not the fundamentals.

4.3. Prices respond to investor credit and the cost of capital

Panel B of Table 3 reports the price equation. We exclude wage growth, which has no direct role in a user-cost price relationship, and we report the regressors both contemporaneously and lagged one quarter, because investor credit and prices are plausibly determined within the same quarter. Investor-credit growth enters positively and significantly in both timings. The cash rate

Table 3: Housing market equations, 2006:Q3–2025:Q4

	(1)	(2)
Panel A. Rent growth equation ($\Delta \ln$ *rent*)		
	FD	MG
Price–rent gap$_{t-1}$	0.001 (0.002)	0.007 (0.005)
Tightness$_{t-1}$	0.006*** (0.001)	0.005*** (0.001)
$\Delta \ln$ WPI	0.849*** (0.165)	0.799*** (0.203)
$\Delta \ln$ Employment	0.155*** (0.039)	0.129*** (0.032)
Observations	370	370
R^2	–	0.570
Panel B. Price growth equation ($\Delta \ln$ *price*)		
	Current (t)	Lagged ($t-1$)
$\Delta \ln$ Investor credit	0.072*** (0.013)	0.050*** (0.013)
Δ Cash rate	0.004 (0.006)	-0.006 (0.004)
Tightness$_{t-1}$	0.003 (0.003)	0.004 (0.003)
Observations	370	370

Notes: Coefficients with Driscoll–Kraay standard errors in parentheses. City fixed effects are used throughout. FD = first-difference estimator; MG = mean-group estimator; Current and Lagged denote contemporaneous and one-quarter-lagged regressors. Dependent variables are $\Delta \ln$ rent (Panel A) and $\Delta \ln$ price (Panel B); wage growth is excluded from Panel B. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

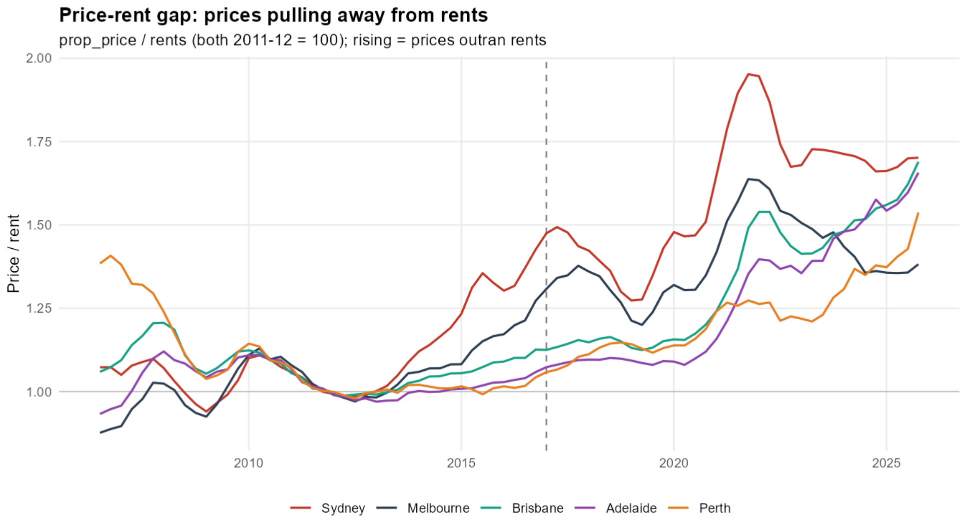

Figure 3: Price–rent gap by capital city: the dwelling price index over the rent index (both 2011–12 = 100). A rising series indicates prices outrunning rents.

is the clearest case where timing matters. Entered contemporaneously it takes a small positive coefficient; entered with a one-quarter lag it turns negative, the sign a user-cost relationship predicts. We read the positive contemporaneous association as the familiar endogeneity of a policy rate that is itself raised during booms, when prices are also rising, and the negative predetermined coefficient as the better-behaved estimate of how the cost of capital feeds into prices. We therefore do not interpret the contemporaneous cash-rate coefficient and treat the lagged specification as the more credible of the two.

4.4. The divergence, and what it is

The descriptive starting point is that prices pulled away from rents over the sample, sharply so from 2020. Figure 3 shows the price–rent ratio rising in every capital, and Figure 4 shows real rents, deflated by wages, moving on a quite different path driven by local conditions rather than by the price cycle. The question is whether this decoupling reflects investor credit operating on prices in a way it does not operate on rents.

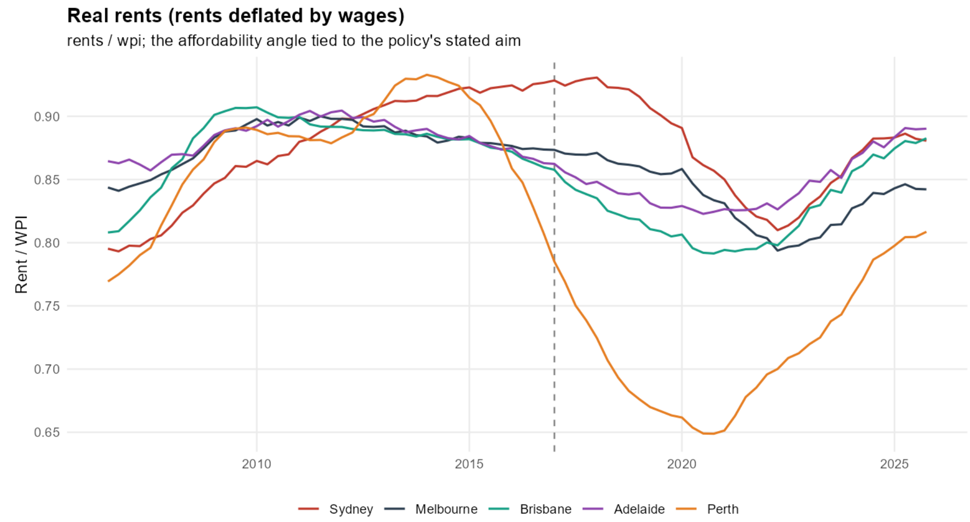

Figure 4: Real rents, rents deflated by the state wage price index, by capital city. Broadly stable-to-declining until the pandemic, then recovering, with Perth's mining-cycle collapse and rebound the main exception.

Table 4: Investor credit × post-2017 interaction, price versus rent

	Price equation	Rent equation
Panel A. Contemporaneous credit		
Δ ln Investor credit × post	0.056*** (0.021)	0.004 (0.008)
Panel B. Predetermined credit (lagged one quarter)		
Δ ln Investor credit$_{t-1}$× post	0.041* (0.023)	0.017*** (0.006)

Each cell is the interaction coefficient from the equation in Table 3 augmented with the interaction. City fixed effects; Driscoll–Kraay standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Contemporaneously the contrast is stark. Interacting investor-credit growth with a post-2017 indicator, the interaction in the price equation is 0.056 ($p < 0.01$) while the same interaction in the rent equation is 0.004 and is not statistically significant ($p = 0.64$), an order of magnitude smaller. Taken at face value this says that after 2017 investor credit became a far stronger correlate of price growth than of rent growth.

We do not take it fully at face value, because contemporaneous investor credit and prices are jointly determined: credit is drawn as purchases settle, so a within-quarter association partly reflects prices acting on credit as much as credit acting on prices. To gauge how much of the contrast is co-movement of this kind, Table 4 re-estimates the interaction with investor credit entered as predetermined, lagged one quarter. Under predetermination the price interaction is 0.041 ($p < 0.10$) while the rent interaction rises to 0.017 and is significant ($p < 0.01$). The two interactions, an order of magnitude apart contemporaneously, converge under predetermination to within roughly a factor of two and a half, as the rent response emerges once credit is lagged.

The reading we adopt is therefore deliberately modest. Investor credit comoves more tightly with prices than with rents, and that asymmetry is large in the same quarter and attenuates, without vanishing, when credit is predetermined. Neither estimate is a structural causal elasticity: the contemporaneous figure overstates the role of credit through simultaneity, and the strictly lagged figure understates it by displacing a relationship that operates largely within the quarter. What survives both is a differential co-movement, prices tracking investor credit more closely than rents do, rather than evidence that credit causally drives prices and leaves rents untouched. The rent equation functions here as a placebo: if the price–credit association were a pure accounting artefact of joint determination, the same force would not so largely spare rents, and the persistence of a small but real credit–rent association under predetermination is consistent with credit affecting rents through a slower, supply-side channel.

We further probe the robustness of the price–credit association in two ways (Appendix C.1).

First, we instrument investor-credit growth with a shift-share (Bartik) instrument formed from each city's pre-2017 investor-credit weight and leave-one-out national credit growth; with only five cross-sectional units the first-stage strength is the binding diagnostic, and we report the IV as indicative rather than as the headline. Second, we confirm that the same instrument does not load on rents, the placebo direction. The qualitative contrast survives, but we lean on the predetermined-regressor and local-projection evidence rather than on the IV.

4.5. The mechanism: investor credit reaches prices fast and rents slowly

To recover the timing that the static interactions in Table 4 compress into a single coefficient, we estimate panel local projections of the price and rent levels on an investor-credit innovation over twelve quarters, with city fixed effects and lagged controls that absorb the predictable component of credit. Figure 5 plots the cumulative response of each log level to a one-standard-deviation credit innovation, with ninety percent Driscoll–Kraay bands. Horizon zero retains the within-quarter simultaneity noted above; horizons from one quarter out are predetermined and so are clean of reverse causality from prices back to credit. The long-horizon estimates are less precise and their windows overlap the pandemic, so we read the figure for its timing rather than for the exact magnitude at distant horizons.

The two responses differ in speed, not in whether credit reaches them. Prices respond on impact, by about 1.2 percent (0.012 log points), build to roughly 2.9 percent (0.029 log points) by the fourth quarter, and remain elevated through the third year, drifting up to about 3.9 percent (0.039 log points) by the twelfth, with a shallow mid-window dip that lies within the bands. Rents are flat and indistinguishable from zero through the first year, then climb steadily to about 2.3 percent (0.023 log points) by the twelfth quarter. The horizon at which the rent response first becomes significant is sensitive to the number of control lags, falling between the second and the seventh quarter depending on that choice, so we treat it as approximate; what is stable across lag specifications is the shape, a delayed rent response that builds over the second and third years, and the twelfth-quarter magnitude of about 0.02 log points. Credit is capitalised into prices almost at once, as an asset-pricing channel implies, and reaches rents only with a lag of around a year or more, at the slower speed of occupancy and supply.

This shape reconciles the static estimates. The contemporaneous interaction reads the impact horizon, where the price response dwarfs the rent response, while the predetermined interaction reads a later horizon, where rents have begun to respond, which is why its rent coefficient is non-zero. Both describe the same paths at different points rather than conflicting results.

Over the full horizon the rent response closes much of the gap with the price response. By twelve quarters the point estimates have narrowed and the confidence bands overlap substantially, so the two responses are no longer statistically distinguishable, though the price response is itself imprecisely estimated that far out. The decoupling documented above is therefore a difference in transmission speed and a medium-run wedge, not a permanent insulation of rents from financial conditions: a credit impulse that lifts prices on impact works through to rents over one to three years. The persistent gap in the descriptive series reflects sustained financial conditions and price-specific drivers such as the cost of capital, rather than rents never responding, and the lag carries a direct affordability implication, since the rent consequences of a credit-driven price boom materialise only after a delay of a year or more.

4.6. The decoupling across cities

The price–rent ratio rose across all five capitals from near parity in the early 2010s to between 1.4 and 2.0 by 2025, most in Sydney (Figure 3). Real rents (rents deflated by wages) were broadly stable-to-declining until the pandemic and then recovered, with Perth the notable exception (a sharp mining-cycle rent collapse and rebound), consistent with rents responding to local fundamentals (Figure 4). We report the cross-city correlation patterns as description only and do not build a cross-sectional gradient on them. The structural break in the investor-credit channel dates to 2017, while its largest movements are concentrated in the pandemic period, so

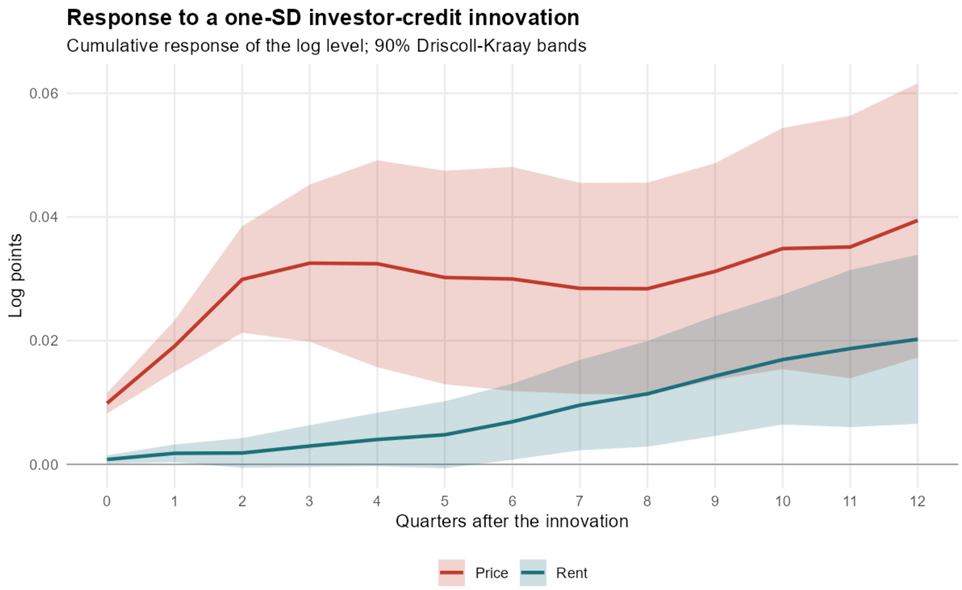

Figure 5: Cumulative response of the price level and the rent level to a one-standard-deviation investor-credit innovation, over twelve quarters, in log points (a response of 0.030 is about 3.0 percent). Panel local projections with city fixed effects and lagged controls; ninety percent Driscoll–Kraay bands. Horizon zero carries the within-quarter simultaneity caveat; horizons from one quarter out are predetermined, and the long-horizon estimates are less precise.

we treat the 2017 package as one marker within a longer structural shift rather than as its cause: the break precedes the pandemic, and the pandemic amplifies it.

5. Discussion

We interpret the results for housing economics, regional analysis, and metropolitan policy.

5.1. Two Markets, Two Sets of Drivers

The central interpretive point is that rents and dwelling prices, which moved together historically, are governed by different forces. Rents track local housing fundamentals, the balance of migration against new completions, real wages, and employment, throughout our sample, with the relative weight on the employment channel rising over time. Prices respond to the cost and availability of capital, and increasingly to investor mortgage demand. The breakdown of the price–rent relationship is not an unexplained correlation but the mechanical consequence of the two outcomes loading on different right-hand sides. When the driver common to both (broad housing demand) ceases to dominate and a price-specific financial driver takes over, the series separate.

This reframes the "decoupling" that single-city studies have documented (Egan and McQuinn, 2023; Greenaway-McGrevy and Phillips, 2023). Rather than a puzzle requiring market-specific explanation, it is the visible surface of a more general fact: rents are the price of a service and remain tied to the fundamentals of that service market, while prices are the valuation of an asset and become tied to the conditions of asset finance. The decoupling is what happens when asset-finance conditions move independently of service-market fundamentals, as they did after 2017 and especially through the pandemic period.

5.2. The Timing and Breadth of the Financialization

Two features of the result discipline its interpretation. The strengthening of the investor channel in prices is dated to 2017 rather than to the pandemic: a separate pandemic interaction adds nothing once a 2017 break is allowed. This matters because the most dramatic movements in our rolling decomposition occur in 2020–2022, and a natural reading would attribute the whole phenomenon to pandemic-era monetary conditions and the associated investor surge. The data do not support that reading; the structural shift in how prices respond to investor credit is in place before the pandemic and is merely amplified by it.

The financialization is also not concentrated in Sydney, the city targeted by the 2017 package. A Sydney-specific investor interaction is in fact negative and significant: the post-2017 price–credit channel operates across the capitals and is, if anything, weaker in Sydney than elsewhere. This is the most important difference between our findings and a policy-evaluation reading of the same period. The change is not a NSW phenomenon transmitted by a NSW reform; if it were, it would be strongest in the treated city, and it is not. It is better understood as a national change in how Australian dwelling prices are determined, of which the NSW package is one contemporaneous event rather than the cause. We therefore resist the temptation, to which a five-city cross-section invites, to build a financialization gradient in which Sydney and Melbourne differ categorically from the smaller capitals. With five units that gradient cannot be identified, and the smaller Sydney response suggests it may not exist in the form earlier work posited.

5.3. International Comparative Perspective

Our mechanism, rents tied to service-market fundamentals and prices increasingly tied to investor finance, generates predictions testable against international evidence. The relevant question abroad is whether the same separation of rent and price drivers appears where investor finance has expanded, with intensities set by local institutional configurations. Canada is the closest analogue, sharing a common national monetary policy across cities of very different financialization intensity; British Columbia's Foreign Buyer Tax (2016) and Ontario's Non-Resident Speculation Tax (2017) are natural experiments akin to the NSW reforms, and Du et al. (2022) find temporary price reductions of 5–9%, concentrated in single-family homes, consistent with our reading that interventions in financialized markets produce short-term effects while deepening segmentation. The framework predicts the sharpest property–rental decoupling in the most heavily invested metropolitan markets and more integrated dynamics elsewhere. Institutional context should mediate its form: where rental supply is dominated by Build-to-Rent landlords pricing on discounted cash flow rather than capital gains, as in parts of the United Kingdom (Hilber and Vermeulen, 2016), decoupling should be weaker; and where public provision is large enough to satisfy demand directly rather than compete with private construction, as in Singapore's HDB system (Phang, 2018), the property–rental link should persist. The weak migration coefficients we observe may partly reflect the absence in Australia of the migration–supply coordination such systems embed. These cases are predictions, not tests; we offer them to indicate that the mechanism is portable rather than to evaluate it abroad.

5.4. Policy Implications

Our central result, that rents remain anchored to fundamentals while prices have migrated onto an investor-finance channel, has a direct implication for affordability policy. Interventions aimed at rental affordability operate on a market still governed by migration, supply, and wages; they are not undone by the financialization of the ownership market, because that financialization has not (or not yet, beyond a small degree) transmitted to rents. Conversely, policies that target the ownership market through investor finance act on the part of the system where the structural change has occurred.

Reforming tax and investor-credit settings. Because the post-2017 strengthening of the price–investor-credit link is broad-based rather than confined to one market, settings that shape investor finance nationally (negative gearing, capital-gains concessions, and macroprudential limits on investor lending) are the natural levers for the price side. This is the margin on which policy has since begun to move. Legislation now before the Parliament, the Treasury Laws Amendment (Tax Reform No. 1) Bill 2026, would limit negative gearing to new builds from the 2027–28 income year: rental losses on established residential properties acquired after 7:30 pm on 12 May 2026 would no longer be deductible against wage and salary income and would instead be quarantined to offset future residential rental income or capital gains, with purchases made before that date grandfathered under the existing rules and eligible new builds, build-to-rent, and social and affordable housing exempt.[1] A companion measure in the same Bill would replace the flat 50% capital-gains discount with cost-base indexation and a 30% minimum tax on net gains accruing from 1 July 2027, reinforcing the same shift. The design logic, phasing in with grandfathering and steering the remaining concession toward new supply, rests on realigning asset returns with rental yields and on tilting investor finance from established stock toward construction. Our evidence speaks to the premise rather than the calibration: that investor credit increasingly drives prices while rents stay tied to fundamentals is consistent with the misalignment such a reform targets having widened after 2017, and with the price side being the appropriate object of a credit- and tax-based instrument. Because the reform's bite would fall on established dwellings purchased after the cutoff and exempt grandfathered holdings and new builds, its near-term effect would operate through a slowly accumulating stock rather than the whole market at once, so any price effect should emerge gradually. We do not estimate the affordability or price effect of the reform and do not claim it would be concentrated in particular cities; our extended panel, which finds the financialization gradient to be broad-based, gives no reason to expect a uniform national measure of this kind to bite unevenly across capitals.

Coordinating migration with housing supply. Migration enters the rent equation through the tightness fundamental, where it is a strong and significant driver: rents respond to the balance of population inflow against new dwelling completions. Because tightness, not migration alone, is what matters, migration policy and construction policy are substitutes for rental pressure; adding dwellings and moderating inflow act on the same ratio. This argues for coordinating migration intake with housing supply capacity rather than treating them separately.

Nonetheless, given the large volume of population flows, coordination with housing supply remains important. Migration policy should align with real-time city-specific metrics: vacancy rates below 2% or rental growth outpacing wages for two consecutive quarters should trigger review mechanisms. Regional incentives (such as accelerated permanent residency in areas with excess capacity) can ease pressure on primary metropolitan areas, and linking international student enrolment to purpose-built accommodation can address concentrated demand in university cities. All measures should be implemented with caution, recognising that the net effect on rental affordability is ambiguous.

Expanding social housing. The 2017 reforms suggest that institutional changes (expanding community housing while reducing public stock) may improve affordability, though the mechanism operates through community housing expansion rather than aggregate supply. Performance-based community housing management could be expanded nationally, with allocations guided by local rental-market conditions, vacancy rates and the balance of migration against new supply, rather than by a presumed ranking of cities by financialization. Rental assistance should likewise be restructured to reflect local market conditions.

Leveraging labour markets. Wages and employment are significant drivers of rents across all five markets, so labour-market policy operates on rental affordability through the fundamentals channel rather than being neutralised by financialization. Training and shared-equity schemes that ease ownership transitions remain useful, but the broader point is that the rental market has not decoupled from wages the way the ownership market has decoupled from rents; wage and employment conditions still move rents. We do not estimate city-specific ownership-transition thresholds and make no claims that the wage–rent relationship reverses across markets.

[1] As at June 2026, the Bill had passed the House of Representatives (4 June 2026) and was before the Senate, having been referred to the Senate Economics Legislation Committee; it was not yet law, and its provisions and commencement remained subject to amendment.

6. Conclusion

Rents and dwelling prices in Australia's five largest capital cities moved together for as long as both were driven by broad housing demand. This paper shows that they have come apart, and why. Estimating two reduced-form equations on the within-city time dimension over 2006:Q3–2025:Q4, we find that rents remain governed by local fundamentals (a migration-to-completions tightness measure, real wages, and employment) throughout the sample, with the weight on the employment channel strengthening over time, while dwelling prices are governed by the cost of capital and, increasingly, by investor mortgage demand.

The central result concerns the differing role of investor credit in the two equations. Investor credit comoves far more tightly with prices than with rents, and this asymmetry widens after 2017. We are careful not to read it as a structural elasticity: because credit and prices are partly determined within the quarter, the contemporaneous association overstates the role of credit while a strictly predetermined one understates it, and the truth lies between. What survives both is a differential co-movement, prices tracking investor credit far more closely than rents do, with the rent equation serving as a placebo against a pure simultaneity artifact.

Panel local projections recover the mechanism behind this contrast. A credit innovation is capitalised into prices on impact, as an asset-pricing channel implies, and reaches rents only with a lag of more than a year, at the slower speed of occupancy and supply; over a three-year horizon the two responses converge. The breakdown of the price–rent relationship is therefore a difference in transmission speed and a medium-run wedge, not a permanent insulation of rents from financial conditions. The persistent gap in the descriptive series reflects sustained financial conditions and price-specific drivers rather than rents never responding.

Two further checks bound the result. The strengthening of the investor channel is not purely a pandemic phenomenon, since a separate pandemic interaction adds little once a 2017 break is allowed, and it is present across all five capitals and not concentrated in Sydney, the city targeted by the 2017 NSW package; if anything it is weaker there. The change is therefore better understood as a general shift in how Australian dwelling prices are determined than as the effect of any single state reform. We are deliberate about scope: the design is descriptive reduced-form, identified off roughly seventy quarters of within-city variation, and we neither estimate a causal policy effect nor build a cross-city financialization gradient that five units cannot support.

For policy, the implication is that affordability interventions act on a rental market still anchored to fundamentals, even as the ownership market decouples from them. Measures that shape investor finance nationally, tax settings and macroprudential limits, operate on the part of the system where the structural change has occurred, while migration, supply, and wage policy continue to move rents through the fundamentals channel.

The mechanism generates predictions beyond Australia: where investor finance has expanded, the same separation of rent and price drivers should appear (Section 5.3). The most valuable extension would be a genuinely powered cross-sectional test of how the strength of this separation varies across markets, which requires sub-metropolitan (SA4/SA3) data; our five-city panel can establish the mechanism in the time dimension but not map its spatial gradient. Household-level data linking tenure, investor ownership, and rents would further test the channel at the level of the individual dwelling.

Funding

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Data Availability Statement

The data and code required to reproduce the results in this paper are openly available in the replication repository at `https://github.com/Ainura-tur/price-rent-disconnect`. The repository contains the full estimation pipeline and the raw input series obtained from the

Australian Bureau of Statistics (under CC BY 4.0), the Reserve Bank of Australia, and the Bank for International Settlements, each redistributed under its source's terms with attribution as documented therein.

Declaration of Interest Statement

The authors report there are no competing interests to declare.

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Appendix A Structural Model

This appendix sets out the supply-and-demand system that the conceptual framework in Section 3.1 summarises. It extends Kuklinski (1970)'s account of regional policy coordination, in which market mechanisms align sectoral policies with regional needs, to a setting where financialization can sever the link between asset prices and the fundamentals of the rental-service market. The body of the paper takes only the reduced form and the four expectations to the data; the full system is recorded here for completeness and is not separately estimated.

Contemporary housing markets operate under dual and sometimes contradictory logics: housing as shelter, which supports labour-market participation and regional productivity, and housing as a financial asset, which attracts speculative investment flows. Total housing supply in city j at time t comprises public provision and private market responses:

$$S_{jt} = S^G_{jt}(\delta_{jt}, \Gamma_{jt}) + S^P_{jt}(C_{j,t-1},\ r_{t-1},\ CO_{j,t-1},\ N_{j,t-1},\ w_{j,t-1},\ M_{j,t-1},\ P_{j,t-1},\ R_{j,t-1}). \tag{6}$$

Public supply S^G_{jt} depends on institutional reforms δ_{jt} and public housing structures Γ_{jt}. Private supply S^P_{jt} depends on construction costs $C_{j,t-1}$ and the national interest rate r_{t-1}, with speculative motives capable of distorting traditional supply responses. Labour-market conditions enter through employment $N_{j,t-1}$ and wages $w_{j,t-1}$; migration $M_{j,t-1}$ influences supply through labour availability and demand expectations; property prices $P_{j,t-1}$ and rents $R_{j,t-1}$ signal profitability to developers.

A key constraint in metropolitan regions is the crowding-out term

$$CO_{j,t-1} = \frac{GFCF^G_{j,t-1}}{\text{Total Construction Capacity}_{j,t-1}},$$

competition between public infrastructure investment $GFCF^G_{j,t-1}$ and housing for limited construction resources.[2] This constraint is most binding in geographically constrained cities. Following Saiz (2010), we classify Sydney as severely constrained (ocean, national parks, the Blue Mountains), Melbourne and Perth as moderately constrained, and Brisbane and Adelaide as less constrained, so the crowding-out coefficient would be largest where construction capacity is least elastic.

Rental demand comprises traditional housing needs Q^{trad} and demand arising from exclusion from ownership Q^{fin}:

$$Q_{jt} = Q^{trad}(R_{jt},\ Y_{j,t-1},\ M_{j,t-1},\ NH_{jt}) + Q^{fin}(P_{j,t-1},\ r_{t-1}),$$

where Q^{fin} reflects demand driven by rising asset prices and declining affordability. Current rent R_{jt} depresses demand contemporaneously; lagged property prices $P_{j,t-1}$ raise rental demand as higher prices push potential buyers into renting; income effects through $Y_{j,t-1}$ are ambiguous, since higher incomes raise housing demand but also enable ownership transitions, with the balance depending on the price-to-income ratio. Interest rates r_{t-1} act through cost-of-living channels, migration $M_{j,t-1}$ augments rental demand directly, and owner-occupier demand NH_{jt} captures substitution between renting and owning.

The equilibrium condition for city j,

$$S^G_{jt}(\delta_{jt}, \Gamma_{jt}) + S^P_{jt}(\cdot) = Q^{trad}_{jt}(\cdot) + Q^{fin}_{jt}(\cdot),$$

[2]Total factor productivity growth from a positive fiscal-spending shock may raise housing demand through incomes (Larsen et al., 2025); we assume its contemporaneous effect on housing-supply productivity is negligible.

can produce a policy misalignment captured by

$$\frac{\partial R_{jt}}{\partial \delta_{jt}} < 0 \quad \text{and} \quad \frac{1}{P_{j,t}}\frac{\partial P_{j,t}}{\partial \delta_{jt}} - \frac{1}{R_{jt}}\frac{\partial R_{jt}}{\partial \delta_{jt}} > 0, \tag{7}$$

that is, institutional reforms and the broader post-2017 environment can widen the gap between property-price growth and rental yields even where rents themselves are little affected, deepening the separation between the asset and service markets. We do not read the first inequality as an estimated causal rent reduction; in our results the determinants of rents are stable and do not shift with the reform. The condition motivates the object we do estimate, whether prices and rents come to load on different drivers, and the reduced-form rent equation (1) follows by solving the system for R_{jt}.

Finally, the system has a regional-performance feedback: when housing costs rise relative to wages, effective labour supply declines,

$$Y_{jt} = A_{jt} \cdot F\left[L_{jt}\left(\frac{w_{jt}}{h_{jt}}\right) \cdot K_{jt}\right],$$

where output Y_{jt} depends on productivity A_{jt}, capital stock K_{jt}, and a labour supply adjusted by the wage-to-housing-cost ratio w_{jt}/h_{jt}. Where housing costs rise fastest relative to wages, effective labour supply is squeezed most, one channel through which the housing-finance nexus feeds back into regional competitiveness.

Appendix B Data Sources and Variable Definitions

Appendix C Statistical Tests and Supplementary Results

C.1 Robustness of the price–credit association

This appendix collects the supplementary evidence on the price–investor-credit relationship referenced in Section 4.4.

Shift-share (Bartik) instrument. We instrument city-level investor-credit growth with a shift-share instrument formed from each city's pre-2017 share of investor credit interacted with leave-one-out national investor-credit growth, in a two-way (city and quarter) fixed-effects specification. The exclusion logic is that national credit conditions, scaled by a city's historical exposure, shift local credit without directly determining local price growth except through credit. The binding caveat is the small cross-section: with five cities the first-stage F is the decisive diagnostic, and where it falls below conventional weak-instrument thresholds we report the IV as indicative only. The placebo direction, the same instrument in the rent equation, does not load, consistent with the credit channel operating on prices rather than being a generic demand artifact. We do not treat the IV as the headline estimate; the predetermined-regressor result in Table 4 and the local projections in Figure 5 carry the argument.

Timing and breadth of the post-2017 shift. Two interaction specifications bound the interpretation of the post-2017 result. Adding a separate post-2020 (pandemic) interaction on investor credit alongside the post-2017 interaction leaves the 2017 term in place while the additional pandemic term is small and imprecise, indicating that the strengthening of the channel is not purely a pandemic phenomenon even though its largest movements occur in 2020–2022. Allowing the post-2017 investor interaction to differ for Sydney, the city targeted by the 2017 package, returns a negative and significant Sydney-specific term: the post-2017 investor channel is present across the capitals and, if anything, weaker in Sydney than elsewhere. Whatever the source of that smaller Sydney response, the shift is plainly not concentrated in the treated market, which is the relevant point for ruling out a Sydney-driven, NSW-policy interpretation. Both readings are consistent with treating the 2017 package as one marker within a longer structural shift rather than as its cause.

Table 5: Data Sources, Variable Definitions, and Coverage

Variable	**Source**	**Geography**	**Coverage**
Outcome and Price Variables			
Rental Price Index	ABS 6401.0 (CPI rent component); monthly CPI indicator for recent quarters	Capital city	1972:Q3–2025:Q4
Property Price Index	BIS (Sydney, all dwellings); ABS 6416.0 spliced to ABS mean dwelling price for other capitals	Capital city	2003:Q3–2025:Q4
Mean Dwelling Price	ABS Total Value of Dwellings (continuation/splice input)	State	2011:Q3–2026:Q1
Construction and Investment			
GFCF Public	ABS State Final Demand (current prices, SA)	State	2003:Q1–2025:Q4
GFCF Private	ABS State Final Demand (current prices, SA)	State	2003:Q1–2025:Q4
Investor Mortgage Demand	ABS Lending Indicators (investor housing commitments, dataflow successor to 5601.0 Table 14)	State	2002:Q3–2026:Q1
National Housing Credit	RBA D2 (investor and owner-occupier credit; net loan-purpose switching)	National	1990:Q1–2025:Q4
Dwelling Completions	ABS 8752.0 (Building Activity)	State	1984:Q3–2025:Q4
Labour Market			
Employment	ABS 6202.0 (Labour Force)	State	1978:Q1–2026:Q2
Wages (WPI)	ABS 6345.0 (Wage Price Index, quarterly)	State	1997:Q3–2025:Q4
Housing Services	ABS State Final Demand (rent and dwelling services consumption)	State	2003:Q1–2025:Q4
Monetary Policy			
Cash Rate	Reserve Bank of Australia	National	1990:Q1–2025:Q4
Demographic			
Net Overseas Migration	ABS 3101.0 (also OMAD by visa group, to 2025:Q3)	State	1981:Q2–2025:Q4
Net Interstate Migration	ABS 3101.0	State	1981:Q2–2025:Q4

Notes: All series quarterly. State-level variables proxy city-level measures where unavailable; capital cities account for 65–79% of state economic activity.

Rolling decomposition. Figure 6 traces the contemporaneous partial R^2 of investor credit in the price equation against its rent-equation counterpart over a twenty-quarter rolling window. The price series rises well above the rent series and peaks in 2021–2022, while the rent series stays low throughout. We report this as a contemporaneous decomposition and therefore subject to the same simultaneity caveat as the contemporaneous interaction; the local projections in the main text are the cleaner dynamic evidence.

Functional form of the price–credit relationship. The linearity of the credit term is not driving the divergence. Re-estimating the price equation with a natural cubic spline in investor-credit growth in place of the linear term leaves the fit essentially unchanged (within-R^2 of 0.218 linear versus 0.219 spline), and a joint test of the spline's curvature components does not reject linearity ($\chi^2_2 = 0.29$, $p = 0.86$). The linear credit coefficient in the price equation is an adequate summary; the price–credit association is not an artefact of imposing a linear form.

Loan-purpose reclassification, 2015–2017. The state-level investor-commitment series spans the 2015–2017 period in which lenders reclassified a material volume of loans between investor and owner-occupier purpose under APRA supervision, which could in principle contaminate the post-2017 interaction. We probe this with the Reserve Bank's national D2 credit series, which is available alongside a measure of the net volume of loans switched between purposes. Three checks point the same way. First, the post-2017 interaction re-estimated on the official national investor-credit series, both raw and adjusted for cumulative switching, is small and statistically insignificant on both prices and rents; the national series carries time variation only and, under city fixed effects, has little independent power, so we read these as consistency checks rather than as substitutes for the cross-city estimate. Second, augmenting the headline price divergence with the contemporaneous switching flow as a direct control over the window in which the flow is observed (from 2015:Q3) leaves the price interaction statistically indistinguishable from its no-control value: the two estimates lie within one pooled standard error, and the switching flow is close to orthogonal to the credit terms (within-window correlations below 0.2 in absolute value). The movement in the point estimate over this truncated window is sampling noise rather than the removal of a reclassification artefact. We conclude that the post-2017 strengthening is not an accounting consequence of the loan-purpose reclassification.

Composition of credit. A complementary national check uses the investor share of housing credit, investor divided by investor-plus-owner-occupier stock, from the D2 series. The share is essentially flat across the break (0.338 before 2017, 0.335 after), so the post-2017 result is not a reflection of a rising aggregate tilt toward investors; the cross-city concentration of investor commitments, not the national composition, is what carries the interaction. Adding the national share to the headline price equation leaves the state-level investor-credit interaction unchanged (0.055, $p < 0.01$), consistent with the cross-city signal being distinct from the aggregate composition.

Inference with five clusters. All of our standard errors, not only the instrumental-variables estimates, inherit the limitation of a five-unit cross-section, and we are explicit about it. A wild cluster bootstrap of the contemporaneous price interaction, resampled at the city level with six-point (Webb) weights appropriate to so few clusters, returns a p-value of about 0.01, corroborating the Driscoll–Kraay inference; with only five clusters we nonetheless read it as supportive rather than as an independent precise figure, since the number of distinct bootstrap statistics is limited. The argument rests on the predetermined-regressor estimate and the local projections, which do not depend on cross-sectional resampling, rather than on bootstrap significance.

Construction of the tightness measure. The rent-equation tightness result is not specific to our preferred construction of the ratio. Replacing total net migration with net overseas migration in the numerator leaves the coefficient essentially unchanged (0.0055 versus 0.0058), and an unsmoothed single-quarter ratio gives a similar positive and significant coefficient (0.0036). The coefficient collapses to insignificance only when the already-stationary ratio is differenced (0.0012, $t = 0.36$), which is the over-differencing the levels treatment is designed to avoid and which confirms that diagnosis rather than undermining the result.

Migration composition. A natural question is whether the tightness–rent relationship operates through rental demand specifically, in which case it should strengthen when the migrant inflow skews toward renters. Using net overseas migration disaggregated by visa group, we form the temporary-visa and student-visa shares of net migration by city (as trailing four-quarter ratios, with the 2020–21 border-closure quarters excluded because total net migration passes through zero and renders the ratio undefined). The share varies nearly as much within a city over time as it does across the panel (within-city standard deviation about 0.96 of the overall), so the test has identifying variation. We find no composition effect: the interaction of lagged tightness with the student share is essentially zero (0.001, $p = 0.82$), the temporary-share version is likewise null, and the same interaction in the price equation, a placebo, is also null. The tightness fundamental drives rents, but how renter-skewed the inflow is does not measurably amplify that relationship. We report this as a null with adequate power rather than as evidence for a composition channel.

Disaggregating the migration flow. The headline tightness ratio sums net overseas and net interstate migration into a single numerator and so forces every migrant to carry the same rent slope. We probe this restriction three ways, none of which displaces the omnibus measure. First, we rebuild tightness from renter-relevant subsets of the flow (overseas only, overseas net of students, and an overseas flow weighted by its temporary-visa share) and horse-race them against the omnibus ratio on a common sample. The omnibus measure fits at least as well as every variant (within-R^2 of 0.624 against 0.51–0.60), so no renter-relevant numerator improves the rent equation. The one notable shift is in magnitude rather than fit: stripping students from the overseas flow roughly doubles its per-head slope (0.012 versus 0.006 for pooled overseas, stable across samples), which says the student component dilutes the average rent slope of overseas migration. This is consistent with the composition result above from the opposite direction: students are the low-slope part of the inflow, even though their share does not modulate tightness. Second, we enter net overseas and net interstate flows with separate slopes. They differ on the full sample, with interstate carrying about twice the overseas slope (0.010 versus 0.005), and a Wald test rejects equality ($\chi^2_1 = 4.08$, $p = 0.04$). But the rejection is not broad-based: leaving out each city in turn, the overseas–interstate contrast strengthens without Sydney or Melbourne, weakens to insignificance without Brisbane or Adelaide, and vanishes entirely without Perth (the contrast flips sign, $t = 0.7$, and the interstate slope collapses onto the overseas one). The wedge is a Perth phenomenon, plausibly its large mining-cycle interstate swings coinciding with sharp rent moves, not a general feature of the capitals; for the panel as a whole the pooled numerator is adequate (Table 6). Third, a price-equation placebo, entering a renter-weighted overseas term in the price equation, is uninformative rather than clean: the price specification will not tolerate any migration regressor without its wage-growth coefficient turning large and negative, so we read neither the contemporaneous nor the predetermined version as evidence on the rent/price separation. Taken together, disaggregating migration does not overturn the single tightness fundamental; it leaves the headline rent equation in place while locating what extra structure exists, the student-dilution magnitude and the Perth-specific interstate slope, as features to note rather than to build on. As with the composition and nonlinearity checks, these inferences inherit the five-cluster limitation, which here is acute because one city carries the only rejection.

Nonlinearity in the tightness–rent relationship. We test whether the rent response to tightness departs from linearity, for example passing through faster in already-tight markets. A natural cubic spline in lagged tightness mildly rejects linearity in the rent equation ($\chi^2_2 = 6.5$, $p = 0.04$), but the departure does not take the convex form that a rental-pressure story predicts: splitting the tightness slope at each city's own median returns an additional high-regime slope that is small, insignificant, and wrong-signed, and the corresponding split in the price equation is the larger of the two. We therefore retain the linear tightness term in the headline specification and make no convexity claim; the nonlinearity the spline detects is minor and not economically interpretable. As elsewhere, these inferences inherit the five-cluster limitation discussed below.

Local-projection lag length. The delayed rent response is robust to the number of control lags in the local projections, though the precise horizon at which it first attains significance is not. Across two, four, and six lags the twelfth-quarter rent response is stable at roughly 0.017 to 0.024 log points and the price response leads it throughout; the first significant rent horizon, by contrast, moves between the second and seventh quarter with the lag count, which is why the main text states that timing as approximate.

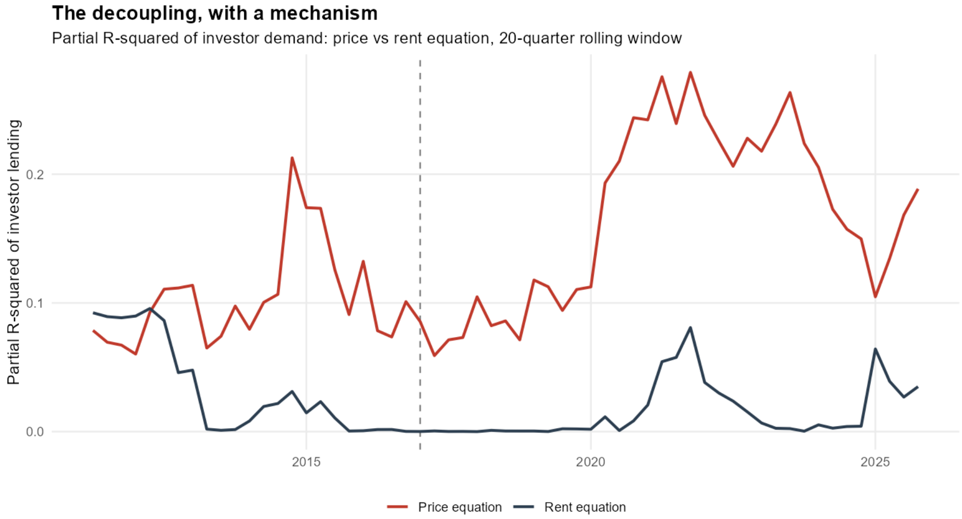

Figure 6: Contemporaneous partial R^2 of investor credit, price equation versus rent equation, twenty-quarter rolling window. Demoted from the main text in favour of the local-projection evidence (Figure 5); reported here as a descriptive complement.

Table 6: Overseas versus interstate migration in the rent equation: leave-one-city-out stability

City dropped	Overseas	Interstate	Difference	t
None (full)	0.0050	0.0098	−0.0048	−2.02
Sydney	0.0040	0.0113	−0.0073	−3.95
Melbourne	0.0050	0.0114	−0.0064	−2.39
Brisbane	0.0050	0.0103	−0.0054	−1.89
Adelaide	0.0050	0.0101	−0.0051	−1.86
Perth	0.0057	0.0046	0.0011	0.74

Notes: Each row re-estimates the rent equation with net overseas and net interstate migration (each scaled by dwelling completions and lagged one quarter) entered as separate regressors, dropping the named city. Columns report the two slopes, their difference (overseas minus interstate), and the t-statistic on that difference under Driscoll–Kraay standard errors. The full-sample rejection of slope equality is not robust: it strengthens without Sydney or Melbourne, weakens below significance without Brisbane or Adelaide, and reverses without Perth, whose interstate slope collapses onto the overseas one. The overseas/interstate wedge is therefore a Perth-specific feature rather than a general property of the capitals, and the omnibus tightness numerator is adequate for the panel as a whole.

Table 7: Unit Root Test Results

	ADF Statistic	
Variable	Level	First Difference
Rental Price Index	-4.40^{***}	-14.01^{***}
Property Price Index	-3.47^{**}	-14.37^{***}
GFCF Public	-2.23	-16.44^{***}
Migration	-4.18^{***}	-12.24^{***}
GFCF Private	-2.26	-14.58^{***}
Housing Services Consumption	-1.42	-13.84^{***}
Total Dwelling Completions	-2.77	-19.72^{***}
Cash Rate	-4.77^{***}	-10.85^{***}
Hourly Wages	-3.95^{**}	-14.33^{***}
Labour Income	-1.87	-13.98^{***}
Full-time Employment	-1.84	-13.89^{***}

Notes: ADF with constant and trend. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. At the five percent level the rental and property price indices, migration, the cash rate, and hourly wages reject a unit root in levels, while public and private GFCF, housing services consumption, dwelling completions, labour income, and employment do not and are integrated of order one. Every series is stationary in first differences. This mixture of integration orders is typical in macro-regional panels and motivates differencing the $I(1)$ series while entering the stationary tightness ratio in levels; we include the lagged price–rent gap as a level control rather than imposing a cointegrating error-correction relationship, which a panel residual test does not robustly support.